

TRONG NHAN NGUYEN

nguyennhan8521@gmail.com — [Website](#) — [GitHub](#) — [ORCID](#)
Busan, South Korea

RESEARCH INTERESTS

Inverse Problems, Image Restoration, Physics-Informed Neural Networks, Computational Imaging

EDUCATION

M.S. in Biomedical Engineering

Pukyong National University, Busan 48513, Korea

Advisor: Prof. [Junghwan Oh](#)

Thesis: [Image Restoration for Scanning Acoustic Microscopy Systems](#)

Relevant Coursework: Advanced Medical Imaging Devices; Computational Polarization Optics; Biosensor and Bioinstrumentation; Smart Signals and Systems in Bioengineering; Nanobioengineering

Oct. 2023 – Aug. 2025

GPA: 4.30/4.50

B.S. in Information Technology Engineering

Ho Chi Minh City University of Technology (HUTECH), Vietnam

Advisor: Prof. [Hung Nguyen](#), MSc. Hoang Khang Vo

Thesis: Software System for Public Administrative Procedure Support

Relevant Coursework: Computer architecture and operating system; Data structure and algorithm; Object-oriented programming; Artificial Intelligence; Software technology

Aug. 2018 – Oct. 2022

GPA: 3.78/4.00

PUBLICATIONS

*Co-First Author, **Co-author

[Google Scholar](#)

Journal and Conference Publications

- [J1] **Trong Nhan Nguyen**, Vu Hoang Minh Doan, Tan Hung Vo, Dang Khoa Pham, Dinh Dat Vu, Thi Thu Ha Vu, Truong Tien Vo, Quoc Dung Nguyen, Jaeyeop Choi, Junghwan Oh. “Unsupervised Hybrid Transformer–Mamba Diffusion for Depth-Dependent Image Denoising in Scanning Acoustic Microscopy,” *IEEE Transactions on Industrial Informatics*, IF = 9.8, 2026. [\[Paper\]](#) [\[Code\]](#)
- [C1] **Trong Nhan Nguyen**, Vu Hoang Minh Doan, Tan Hung Vo, Jaeyeop Choi, Junghwan Oh. “Transformer-Based Super-Resolution Scanning Acoustic Imaging for Industrial Inspection,” *IEEE International Conference on Industrial Technology (ICIT 2025)*, 2025. [\[Paper\]](#)
- [J2] **Trong Nhan Nguyen**, Vu Hoang Minh Doan, Tan Hung Vo, Dinh Dat Vu, Le Hai Tran, Thi Thu Ha Vu, Truong Tien Vo, Quoc Dung Nguyen, Sudip Mondal, Jaeyeop Choi, Byeongil Lee, Junghwan Oh. “Image-enhanced Generative Model for Industrial Scanning Acoustic Super-resolution in Non-destructive Testing,” *Engineering Applications of Artificial Intelligence*, IF = 9.0, 2025. [\[Paper\]](#) [\[Code\]](#)
- [J3] **Trong Nhan Nguyen**, Vu Hoang Minh Doan, Tan Hung Vo, Jaeyeop Choi, Junghwan Oh. “DIPSAM: Diffusion inpainting for restoring water-loss degradation in scanning acoustic microscopy,” *Computers & Electrical Engineering*, IF = 5.5, 2026. [\[Paper\]](#) [\[Code\]](#)
- [J4] Minh Ngoc Nguyen, Khai Le-Duc, Tan-Hanh Pham, **Trong Nhan Nguyen****, Trang Nguyen, Quang Minh Luu, Ba Kien Tran, Truong-Son Hy, Viktor Dremine, Sergei Sokolovsky, Edik Rafailov. “A Wearable Device Dataset for Mental Health Assessment Using Laser Doppler Flowmetry and Fluorescence Spectroscopy Sensors,” *Communications Medicine (Nature Portfolio)*, IF = 7.4, 2026. [\[Paper\]](#) [\[Code\]](#)
- [J5] Thi Thu Ha Vu, Tan Hung Vo, **Trong Nhan Nguyen***, Jaeyeop Choi, Le Hai Tran, Vu Hoang Minh Doan, Van Bang Nguyen, Wonjo Lee, Sudip Mondal, Junghwan Oh. “GAN-Based Super-Resolution in Linear R-SAM Imaging for Enhanced Non-Destructive Semiconductor Measurement,” *Applied Sciences*, IF = 2.5, 2025. [\[Paper\]](#)
- [J6] Truong Tien Vo, Quy Phuong Le, **Trong Nhan Nguyen****, Sudip Mondal, Jaeyeop Choi, Byeongil Lee, Junghwan Oh. “A Non-Contact Ballistocardiogram-Based System with Multi-Task Deep Learning for Real-Time Vital Signs Monitoring in Acupuncture,” *Computers in Biology and Medicine*, IF = 6.3, 2026. [\[Paper\]](#)
- [J7] Thi Thu Ha Vu, Tan Hung Vo, **Trong Nhan Nguyen****, Jaeyeop Choi, Sudip Mondal, Junghwan Oh. “Optimizing Scanning Acoustic Tomography Image Segmentation With Segment Anything Model for Semiconductor Devices,” *IEEE Transactions on Semiconductor Manufacturing*, IF = 2.3, 2024. [\[Paper\]](#)
- [J8] Truong Tien Vo, Huu Sang Nguyen, **Trong Nhan Nguyen****, Le Hai Tran, Quy Phuong Le, Tan Tien Nguyen, Jaeyeop Choi, Junghwan Oh. “Multimodal Smart Clothing with Haptic Feedback for Real-Time Muscle Activation Assessment in Self-Coaching Fitness,” *IEEE Internet of Things Journal*, IF = 8.9. [\[Paper\]](#)

Manuscripts Under Review

- [C2] **Trong Nhan Nguyen**, Junghwan Oh, Truong Son Hy. “WaFIR: Wavelet-Fourier Mixture-of-Experts with Subband-Adaptive Routing for Blind All-in-One Image Restoration,” AAAI Conference on Artificial Intelligence (AAAI), 2027. [\[Code\]](#)
- [J9] Vu Hoang Minh Doan, **Trong Nhan Nguyen****, Thanh Dat Le, Junghwan Oh. “Effective Augmentation Diversity Predicts Template Element Replacement Effectiveness in XRD-Based Crystal Phase Classification,” *Journal of Chemical Information and Modeling*, 2026.

RESEARCH EXPERIENCE

Pukyong National University, Busan, Korea

Supervisor: Prof. Junghwan Oh

Graduate Research Assistant

Oct. 2023 – Present

[1] Project: Automated Robotic Inspection of Weld Surfaces - [Code]

- Led a small team (3 graduate students), planned the project timeline, coordinated daily tasks, and reported weekly progress to the professor.
- Integrated a Yaskawa robotic arm, a Micro-Epsilon laser scanner, and a Basler camera into an automated pipeline for weld identification and 3D surface scanning.
- Achieved 95.36% weld-identification accuracy with the vision module.
- Developed a real-time robot control system for laser surface scanning at 65 mm/s with $100\text{ }\mu\text{m} \times 49\text{ }\mu\text{m}$ resolution, enabling precise 3D visualization of weld surfaces.

[2] Project: AI-Based Defect Inspection for Scanning Acoustic Microscopy - [Code]

- Developed a U-Net segmentation model with a ResNet50 encoder to segment solder voids and chip regions in SAM images of power semiconductor modules.
- Built a sliding-window inference pipeline that segments full-resolution scans.
- Deployed the pipeline as the automated defect-inspection function in the software of [Ohlabs' commercial SAM systems](#), now used by industrial customers.

[3] Project: AI-Based MLCC Inspection System - Client: [Samwha](#)

- Developed an image-processing pipeline to detect abnormal multilayer ceramic capacitors (MLCCs).
- Integrated the detected defect positions with an automated laser-marking stage.
- Trained a YOLO-based detector to verify that defective MLCCs were correctly laser-marked, covering five MLCC sizes (1005, 1608, 2012, 3216, 3225).
- Deployed the system at the customer's production site, where it remains in use and reduces the time and manual effort needed for defect detection and post-marking inspection.

[4] Project: Image Restoration for Ultrasound Imaging - Papers: [SAMGAN](#), [DIPSAM](#), [SAMID](#)

- Started the lab's research line on AI-based SAM image restoration, which led to follow-up studies by labmates, including GAN-based super-resolution for linear R-SAM ([LinearTGAN](#), *Applied Sciences*, 2025) and an M.S. thesis on out-of-focus restoration (N. D. Quan, Pukyong National University, 2026).
- Super-resolution: developed a GAN-based model that reconstructs high-resolution images from coarse-step scans, reducing scanning time by up to 47.2%; integrated as an optional function in the lab's scanning software.
- Inpainting: developed a latent diffusion framework, trained on synthetic masks, that restores regions lost to water-coupling failure and improves downstream defect segmentation.
- Denoising: developed a hybrid Transformer–Mamba diffusion model with a physics-guided synthetic data pipeline (degradation modeling and CycleGAN domain transfer) to remove depth-dependent noise without clean references.
- Published four first-author papers: IEEE Transactions on Industrial Informatics, Engineering Applications of Artificial Intelligence, Computers & Electrical Engineering, and IEEE ICIT 2025.

University of Alabama at Birmingham, AL, USA

Supervisor: Prof. Truong-Son Hy

Research Collaborator

Nov. 2025 – Present

[1] Project: Frequency-Space Learning for Blind Image Restoration - [Code]

- Proposed WaFIR, a Wavelet–Fourier mixture-of-experts with subband-adaptive routing that derives frequency priors from the input image to guide blind all-in-one restoration under unknown degradations (noise, haze, rain).
- First-author manuscript under review at AAAI 2027.

[2] Project: Mental Health Assessment study - [Code]

In collaboration with researchers from Harvard Medical School (USA), the University of Toronto (Canada), and Aston University (United Kingdom)

- Ran classification models on the dataset to establish baseline results for the mental health assessment task.
- Performed data analysis and prepared the figures for the manuscript.
- Revised the manuscript and drafted responses to the reviewers' comments during peer review.

Artificial Intelligence Center, FPT Software Company, Vietnam

Supervisor: Dr. Duc-Vinh Nguyen

AI Engineer

Oct. 2022 – Dec. 2023

[1] Project: Machine Learning for Industrial Process Optimization - Client: [Ameresco](#)

- Analyzed environmental and operational data from multiple museum facilities in Power BI and built a master table.
- Developed machine learning models for univariate CO₂ forecasting and for clustering facility operating patterns.
- Designed data-driven algorithms to estimate heat gain under varying CO₂ levels, optimized the prediction pipeline, and deployed the model to edge devices via Microsoft Azure.

[2] Project: Root-Cause Analysis in Steel Production - Client: [Yamato Kogyo](#)

- Studied the steel production process, built a master table of process data, and analyzed all features in Power BI.
- Applied feature extraction and feature importance to identify the process variables linked to production defects.
- Used Fishbone (Ishikawa) diagrams to determine and document the root causes.

[3] Project: Deep Learning for Search Engine Enhancement - Client: [FPT Retail](#)

- Developed a hybrid search pipeline that combines full-text and semantic search.
- Evaluated retrieval performance of the hybrid pipeline against full-text search alone.

[1] Project: Vietnamese Chatbot and OCR for Public Administrative Procedures - [Code]

- Developed an OCR-based system that extracts personal information from identity documents and automatically fills in administrative forms.
- Collaborated with a team member to build a Vietnamese chatbot that guides citizens through public administrative procedures and auto-fills forms.

TALKS & PRESENTATIONS

IEEE International Conference on Industrial Technology (ICIT) 2025

Presented “Transformer-Based Super-Resolution Scanning Acoustic Imaging for Industrial Inspection.”

Research Seminar, Brain Korea 21 (BK21) Program 2026

Presented the research summary report, which covered ongoing research progress in AI-based industrial inspection and image processing.

HONORS, AWARDS, AND ACHIEVEMENTS

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- Pukyong National University Global Challenge Scholarship 2025
 - Blue Scholarship, Brain Korea 21 (BK21), Korean Government Scholarship Program 2025 & 2026
 - Fire Grant Scholarship 2025 & 2026
 - Scholarship, Pukyong National University 2023
 - Impressive Award, HUTECH Start-up Wing 2021
 - Consolation Prize, AI Hackathon Contest 2020
 - Rewards of Undergraduate Research, Ho Chi Minh City University of Technology (HUTECH) 2020
 - Outstanding Undergraduate Student Scholarship, Ho Chi Minh City University of Technology (HUTECH) 2018

TECHNICAL SKILLS

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- **Machine Learning:** Diffusion Models, Transformers, Mamba / State-Space Models, GANs, CNNs; Image Restoration (denoising, super-resolution, inpainting), Segmentation.
 - **Frameworks & Libraries:** PyTorch, Hugging Face (Transformers, Diffusers, Accelerate), Weights & Biases, OpenCV, scikit-learn, NumPy, Pandas, Matplotlib.
 - **Infrastructure & Tools:** CUDA, Multi-GPU/ Distributed Training, Slurm, Linux, Git, Docker.
 - **Programming Languages:** Python, C.
 - **Instrumentation & Systems:** Scanning Acoustic Microscopy (NDT), Basler cameras, Micro-Epsilon laser sensors, Yaskawa robotic arms, 3D scanners, Ultrasound Imaging System.
 - **English Proficiency:** IELTS 7.5

SERVICE & PROFESSIONAL DEVELOPMENT

Peer Reviewer: Scientific Reports; IEEE Sensors; IEEE International Conference on Industrial Technology; Machine Vision and Applications; Discover Artificial Intelligence; Discover Sensors.**Research Mentor:** Amity Science Research Program, Amity Regional High School, CT, USA (2026 – Present); remotely guiding a high-school student through a year-long independent research project on acoustic imaging and data analysis.**Research Collaborations:** North Carolina A&T State University ([Dang Khoa Pham](#)); Harvard Medical School ([Dr. Tan-Hanh Pham](#)); Aston University ([Dr. Minh Ngoc Nguyen](#)); University of Toronto ([Khai Le-Duc](#)); University of Alabama at Birmingham ([Prof. Truong-Son Hy](#)).**REFERENCES**

Prof. Junghwan Oh jungoh@pknu.ac.kr
Professor of Biomedical Engineering, Pukyong National University, Korea +82 51-629-5771
Relationship: Supervisor**Prof. Truong-Son Hy** thy@uab.edu
Assistant Professor of Computer Science, University of Alabama at Birmingham, USA +1 205-934-8604
Relationship: Research Collaborator**Prof. Sudip Mondal** smondal@pknu.ac.kr
Associate Professor, Digital Healthcare Research Center, Pukyong National University, Korea +82 51-629-7860
Relationship: Lab Co-supervisor